Numerical Companion to "Rollout Gates and Privileged Distillation for Lane Selection with Sampled Safety: When Is Run-Time Simulation Enough?"

The experiments of the paper: what each runs, what it compares, how each reported number is computed, and what it means

Hossein Nick Zinat Matin*, Jeffrey Sheng, Xiaoqian Gong, and Jesse Read†

Version 11, 7 November 2026 (release accompanying the conference submission)

*H. Nick Zinat Matin and J. Read are with the Laboratory of Computer Science (LIX), Ecole Polytechnique, Palaiseau, France. J. Sheng and X. Gong are with the Department of Mathematics, Amherst College, MA, USA. hossein.matin@polytechnique.edu, jsheng26@amherst.edu, xgong@amherst.edu, jesse.read@polytechnique.edu. *Corresponding author.

†H. Nick Zinat Matin acknowledges that this work benefited from Hi! PARIS and State funding managed by ANR under the France 2030 program, reference ANR-23-IACL-0005.

Summary. One longitudinal controller (PPO) drives inside a lane, always through a safety filter. A critic (Double DQN) proposes lane changes. The paper asks *which proposals to execute* and, more precisely, *when a run-time simulation of the proposal is enough and when a learned rule is needed.* The ideal answer is the **privileged gate**: simulate "change now, then follow the critic" against "stay, then never change" with the true hidden traffic state and execute the change only if it is better. A **prior-sampling gate** does the same with the hidden state drawn from a heuristic law built on the declared ranges (deployable, Monte-Carlo; Section 5). The **student** is a small network trained on the privileged gate's labels, so it decides in 0.3 ms. The **cascade** uses the student where it is confident and queries the sampler near ties. The paper's main experiments vary the uncertainty level ν of the traffic, the query delay, and the information given to the sampler, and certify the deployed rules.

Two versions. In *version D* (deterministic traffic, packages A/B/C) "exact" means one simulation. In *version S* (stochastic traffic with hidden parameters, packages S, T, V, W, W2, X, Y) a simulation is a noisy sample, and whether learning or sampling wins depends on what is hidden and on the delay.

How to read this note. Sections 1–7 describe the system and the rules . Sections 8–17 describe each experiment of the paper in order, the exact quantity each number is, the result tables behind the paper's numbers (per-seed values are listed where a decision depends on them, as for certification), and the interpretation. Package letters name batches of runs; Section 8 maps each to the paper. **Code.** The code is released with this note at `https://github.com/hosein-matin/Lane-change-switch-RL` (code under the PolyForm Strict License 1.0.0; this note and the data exports under CC BY-NC-ND 4.0): one runner folder per package family (`deterministic/`: packages A–C; `stochastic_S/`, `stochastic_T/`; `stochastic_V/`: packages V, W, W2, X, Y, Z and the benchmark) and `analysis/`, whose README lists the script behind each number. File names such as `aggV/...` and `bench/...` in this note are outputs in the analysis working folder; the summary exports behind the deterministic tables (`summary_final.json`), packages S and T including the label budget (`summary_S.json`), the sweep (`summary_V.json`, `value_of_sim.json`, `decomp.json`) and packages X, Y and Z are released in `data/`.

1 Notation

Object	Meaning (values used)
Δ, T_m, T_f	sample time 0.1 s; lane-change commitment 1 s ($N = 10$ samples); episode 15 s
L, q	number of lanes (2 or 3); current lane
actions	0 = stay (Δ), 1 = change left, 2 = change right (each T_m); with two lanes one direction never exists
$\mathcal{A}(y)$	admissible actions: stay iff $I_q^{\Delta} \neq \emptyset$; change iff the target slot exists, the entry guard holds, $\eta \geq \tau_d = 2$ s, and $t + T_m \leq T_f$
I_q^{Δ}, $I_q^{T_m}$	held-input intervals (Prop. 1): any constant input in them keeps clearances $\geq d_0 = 2$ m and $0 \leq v \leq v_{\max}$ for Δ (resp. T_m)
entry guard	$I_q^{T_m} \cap I_p^{T_m} \neq \emptyset$ (old lane q, new lane p): one input safe in both lanes for the whole commitment
c_0, $\bar{c}_0$	lane-change cost in the reward (2); training-time cost of the critic (grid, validated)
π_b	critic policy: masked argmax of Q_ω, ties to stay
π_s	"never change": stay whenever staying is admissible, otherwise act as π_b
proposal epoch	a decision at which π_b proposes a change and staying is admissible
$g(y)$	advantage of the proposal over never changing: $Q_b(y, \mathsf{c}) - V_s(y)$
$\hat{g}$	an estimate of g (forecast rollouts, Monte Carlo, or the student)
charge	loss of a wrong decision: $\lvert g \rvert$ (accept when $g < 0$, or reject when $g \geq 0$)

2 The setup in full

2.1 Model, filter, reward

Item	Value
vehicles	ego + one leader and one follower per lane: 5 (two lanes) or 7 (three lanes); length 5 m
ego	$\dot{x} = v$, $\dot{v} = u$, $u \in [-6, 2.5]\,\mathrm{m/s^2}$, $0 \le v \le 20\,\mathrm{m/s}$; exact held-input update
followers	optimal velocity model $\dot{v} = \alpha(V(d) - v)$, $d_c = 15\,\mathrm{m}$, $b = 8\,\mathrm{m}$, desired speed $U(15, 20)\,\mathrm{m/s}$, saturation $[-7, 2.5]\,\mathrm{m/s^2}$, 10 substeps per sample; version D: $\alpha = 0.8$; version S: $\alpha \sim U(0.5, 1.1)$ hidden
leaders	version D: $\bar{v} + A\sin(\omega t + \phi)$, $A \le 5\,\mathrm{m/s}$, $\omega \in [0.2, 0.8]\,\mathrm{rad/s}$ (per family in Table 1); version S: random target tracking (Section 5)
declared bounds	leader $\ge -A\lvert\omega\rvert$; follower: monotone envelope of the follower model over the certified window T: the gap to the ego is at most $d^{\uparrow} = d + (v - v_f)_+ T + \frac{1}{2}(\overline{u} + b_f)T^2$ for every admissible ego input sequence, the follower's speed is at least $v^{\downarrow} = v_f(1 - \alpha\Delta/10)^{10T/\Delta}$ (as $\dot{v}_f \ge -\alpha v_f$), and V is nondecreasing, so $\overline{a}^- = \mathrm{clip}\left(\alpha(V(d^{\uparrow}) - v^{\downarrow}); [-7, 2.5]\right)$ bounds its acceleration whatever the ego does; both saturations are needed, since without the lower one the expression can fall below the follower's actual deceleration -7 and stop being a bound (version S: worst case over the declared ranges, 7 gains and the largest desired speed; the unclipped expression is convex in the gain on the declared ranges, and the monotone clip keeps the maximum at an endpoint of the gain range). The simulator has always used this clipped form (`env.py`, `follower_bound`); version 9 of this note printed the expression without the lower saturation
filter	$d_0 = 2\,\mathrm{m}$; executed $u = \mathrm{clip}(u_c; I_q^{\Delta} \cap I_q)$, or $\mathrm{clip}(u_c; I_q^{\Delta})$ if the intersection is empty, where I_q^{Δ} is the sampled certificate of Proposition 1 and I_q the sample-time barrier interval $\{u : \underline{u} \le u \le \overline{u},\ -\kappa_v v \le u \le \kappa_v(v_{\max} - v),\ u \le \underline{a}^+ + (\lambda + \kappa)D^+ + \kappa\lambda H^+,\ u \ge \overline{a}^- - (\lambda + \kappa)D^- - \kappa\lambda H^-\}$ with $\lambda = 0.6$, $\kappa = 1$, $\kappa_v = 2\,\mathrm{s^{-1}}$. I_q only narrows the interval; safety rests on I_q^{Δ}. In package X the kernel bound of Theorem 1 is applied after this clip and takes precedence
commitment	ego counted in both lanes for $T_m = 1\,\mathrm{s}$; one input from $I_q^{T_m} \cap I_p^{T_m}$ held
reward	per sample $w_d\Delta x - w_u u^2 \Delta$ ($w_d = w_u = 1$); $-c_0 = -2$ per lane change; $-C_{\mathrm{fail}} = -200$ if no action is admissible; undiscounted, $T_f = 15\,\mathrm{s}$

2.2 Learning

Component	Details
longitudinal controller (PPO)	actor and value 2×64 tanh; 10 inputs (speed, front/rear clearance and relative speed, declared bounds, filter interval, remaining time); Gaussian action mapped to $[-6, 2.5]$; 500 iterations $\times$ 64 fixed-lane episodes; $\gamma = 1$, GAE 0.95, clip 0.2, 8 epochs, 8 minibatches, learning rates $3 \cdot 10^{-4}/10^{-3}$; trained once (two-lane episodes), reused everywhere
critic (Double DQN)	2×128 tanh, 2 outputs (two lanes, version D) or 3 (three lanes and version S); 6400 episodes in batches of 32; replay $3 \cdot 10^5$; batch 256; learning rate $3 \cdot 10^{-4}$; target copy every 1000 steps; ϵ from 1 to 0.05 over the first half; rewards $\times 0.1$; best network on a fixed 96-scenario validation set kept
critic inputs	time left, speed, cooldown; per lane (current, other / current, left, right): front and rear clearance and relative speed, declared bounds, slot validity; no disturbance parameters. Version S "critic + acc" adds the measured leader accelerations
training cost $\bar{c}_0$	version D: grid $\{2, 5, 7, 10, 14, 20\}$ (three lanes $\{2, 5, 10, 14, 20\}$), chosen per seed on 300 validation scenarios; version S: $\bar{c}_0 \in \{2, 10\}$ (10 = typical validated value)
student	2×128 tanh regression; inputs: critic features + measured leader accelerations (+ direction); version D: Huber loss on g clipped to ± 30, 300 epochs; version S: squared error on $\bar{g}$ clipped to ± 60, 200 epochs; 20% held out; threshold τ on validation
router (version S)	same network shape, 4 outputs (charges of accept, reject, query; work of a query), squared error

2.3 Scenarios and statistics

Seven families, 100 held-out scenarios each per seed; the ego starts in a random lane. Two lanes: *random*; *equal lanes* (identical traffic, a change can only cost); *faster target* (target leader 3–6 m/s faster, large gaps); *fast target follower* (faster target, but a close fast follower there); *leader braking* (own leader brakes strongly in the first seconds); *front/rear conflict* (close own leader that brakes, close fast own follower); *late advantage* (target lane becomes faster only after ≈ 10 s). Three lanes: random, equal, faster adjacent, *faster two lanes away* (behind a slower middle lane), leader braking, front/rear conflict, late advantage. Training uses a mixture (40% random, 10% each other family) with separate seeds; validation 300 mixture scenarios; the $n = 500$ calibration scenarios of each seed are drawn from the same mixture with the stream $123\,000 +$ seed; tests are used once. Every initial state is drawn from its family and rejected and redrawn unless it lies in the admissible domain (nonnegative excess gaps $H \geq 0$ and $\psi \geq 0$, speeds within bounds, nonempty I^{Δ}); rejections are counted by the sampler (`scenarios.py`). Seeds: version D 15 (two lanes) and 10 (three lanes); version S 10 each. Statistics: paired differences to the critic of the same seed; every interval in the paper and in Sections 8–14 is a 95% t-interval over seeds. The version-D tables of earlier packages used a two-way bootstrap (seeds and scenarios), and the package-T table in Section 16 reports one standard error; their $\pm$ values are therefore narrower than a 95% t-interval and are labelled as such.

Initial states are redrawn until both lanes' guards hold. In version S the same parameters define each leader's pattern $\bar{v} + A\sin(\omega t + \phi)$ around which its piecewise-constant targets are drawn (target noise $0.6\nu A\mathcal{U}(-1, 1)$, hard-braking target $\bar{v} - 1.5A - 2$ with probability $\min\{0.2\nu, 0.6\}$, segment lengths $\mathcal{U}(0.5, 3)$ s). Test scenarios are generated from fixed seeds per family and are identical for all rules and seeds.

Table 1: Scenario generators (version D and the leader patterns of version S). $\mathcal{U}(a,b)$ uniform; $v_e \sim \mathcal{U}(8,15)$ is the ego speed; "c" the ego's lane, "t" the adjacent (two lanes) or each other lane (three lanes); d^+, d^- initial front and rear gaps, v_f follower speed; leader $\bar{v} + A\sin(\omega t + \phi)$ with $\bar{v}$ clipped to $[4,19]$, $A \le \min\{\bar{v}, 20-\bar{v}, A_{\text{cap}}\}$ ($A_{\text{cap}} = 3$ unless stated), $\omega \sim \mathcal{U}(0.2,0.7)$, $\phi \sim \mathcal{U}(0,2\pi)$ unless stated; follower desired speed $\mathcal{U}(15,20)$. Three-lane families use the same blocks per lane.

family	parameters
random	$\bar{v}_c, \bar{v}_t \sim \mathcal{U}(6,18)$; $d_c^+ \sim \mathcal{U}(6,45)$, $d_c^- \sim \mathcal{U}(6,40)$, $v_{f,c} \sim \mathcal{U}(6,16)$; $d_t^+ \sim \mathcal{U}(3,50)$, $d_t^- \sim \mathcal{U}(3,40)$, $v_{f,t} \sim \mathcal{U}(6,18)$
equal	one leader $\bar{v} \sim \mathcal{U}(8,16)$ copied to both lanes; $d^+ \sim \mathcal{U}(10,40)$, $d^- \sim \mathcal{U}(10,35)$, $v_f = v_e + \mathcal{U}(-1,1)$, identical in both lanes
faster target (adjacent)	$\bar{v}_c \sim \mathcal{U}(7,12)$, $\bar{v}_t = \bar{v}_c + \mathcal{U}(3,6)$, $A_{\text{cap}} = 1$; $d_c^\pm \sim \mathcal{U}(10,30)$; $d_t^+ \sim \mathcal{U}(25,50)$, $d_t^- \sim \mathcal{U}(20,40)$; $v_f = v_e + \mathcal{U}(-1,1)$
fast follower (two lanes)	as faster target, but $d_t^- \sim \mathcal{U}(5,14)$ and $v_{f,t} = v_e + \mathcal{U}(2,5)$, follower desired speed $\ge v_{f,t} + 1$
faster far (three lanes)	ego in an outer lane; far lane $\bar{v} + \mathcal{U}(4,7)$ with $d^+ \sim \mathcal{U}(25,50)$, middle lane $\bar{v} - \mathcal{U}(0,1.5)$, own lane $\bar{v} \sim \mathcal{U}(8,12)$; $A_{\text{cap}} = 1$
leader braking	$\bar{v}_c \sim \mathcal{U}(9,14)$, $A_c \sim \mathcal{U}(3,5)$, $\omega \sim \mathcal{U}(0.5,0.8)$, $\phi \sim \mathcal{U}(0,1.2)$ (decelerating within ≈ 2 s); other lanes $\bar{v}_c + \mathcal{U}(-1,2)$, $A_{\text{cap}} = 1$; $d_c^\pm \sim \mathcal{U}(12,30)$; $d_t^+ \sim \mathcal{U}(15,45)$, $d_t^- \sim \mathcal{U}(10,35)$
squeeze	as leader braking, with $d_c^+ \sim \mathcal{U}(10,22)$, $d_c^- \sim \mathcal{U}(5,10)$, $v_{f,c} = v_e + \mathcal{U}(0.5,2.5)$, $d_t^- \sim \mathcal{U}(8,30)$
late	$\bar{v}_c \sim \mathcal{U}(9,13)$, $A_{\text{cap}} = 0.8$; target $\bar{v}_c + \mathcal{U}(0,1.5)$, $A \sim \mathcal{U}(3,4)$, $\omega = \pi/\mathcal{U}(11,14)$, $\phi = -\pi/2$ (slowest at $t = 0$, fastest at $t \approx 11$–14 s); $d_c^\pm \sim \mathcal{U}(12,30)$, $d_t^+ \sim \mathcal{U}(15,35)$, $d_t^- \sim \mathcal{U}(12,30)$

3 What both versions share

1. **Safety filter, every sample, everywhere.** The executed input is $u = \text{clip}(u_c; I_q^\Delta \cap I_q)$ (or I_q^Δ alone if the intersection is empty; I_q is defined in the table above). During a lane change the *commitment model* counts the ego in both lanes, and one input from $I_q^{T_m} \cap I_p^{T_m}$, chosen at the first sample, is held for the whole second. This is the only source of the clearance guarantee; no learned component can override it.

2. **Longitudinal controller.** PPO, 500 iterations of 64 episodes inside a fixed lane, executing through the filter; then frozen. The same network is used during stays and lane changes. Package S reuses the controllers trained earlier.

3. **Critic.** Double DQN (2×128 tanh), 6400 episodes, exploring only admissible actions. Its input describes clearances, relative speeds, declared bounds, time, and cooldown in the current, left, and right lanes. It does *not* see the leaders' disturbance parameters (amplitude, frequency, phase).

4. **Proposal and advantage.** At a proposal epoch two continuations are compared, both to the end of the episode: (a) change now, then π_b; (b) stay now, then π_s (never change). The difference of returns is g. Two continuations are compared, not one step, so a change that pays off only later, or only after a second change, is valued correctly.

Why this way

Comparing against "never change" (not against "stay one step, then π_b") is what makes the guarantee work: a gate that follows the exact sign of g is never worse than the better of π_b and π_s, and any other decision rule loses at most the charges $|g_k|$ of the decisions in which it departs from that sign (Proposition 3 of the paper). The deterministic gate of Algorithm 1 adds a failure veto; a vetoed proposal with $g > 0$ is such a departure and is charged. A change whose rollout fails carries $-C_{\text{fail}} = -200$, so $g \ge 0$ with a failed change branch requires the stay branch to be at least as bad; the veto therefore acts only in such rare cases.

4 Version D: deterministic declared model (packages A, B, C)

4.1 Traffic

Leaders follow $v(t) = \bar{v} + A\sin(\omega t + \phi)$; followers follow the optimal velocity model with fixed parameters. Given the scenario, the future is determined. The declared bounds are $-A|\omega|$ for leaders and a bound derived from the follower model over the certified window.

4.2 Exact gate (the teacher)

Algorithm 1 Exact gate at a proposal epoch k (version D)

1: clone the state y_k twice in the *training simulator* (exact traffic)
2: branch C: force the proposed change, then follow π_b to T_f; return G_C (failure: $-C_{\text{fail}}$)
3: branch S: force stay, then follow π_s (never change) to T_f; return G_S
4: $g \leftarrow G_C - G_S$
5: **if** $g \geq 0$ and branch C did not fail **then** execute the change **else** stay ▷ failure veto: not the pure sign rule

Cost: two rollouts of up to 150 samples; 665 ms (two lanes) or 640 ms (three lanes) at the episode start on one laptop core. It is used to *train* the student, as a *reference row* in the tables, and to *compute certificates* offline. In this Python implementation it does not run within a control step, and in any implementation it uses the simulator's knowledge of the traffic, so it is not deployable (compiled timings of the stochastic query are in Section 11).

4.3 Forecast gate (deployable simulation)

The same algorithm with a forecast instead of the simulator: leaders keep their measured acceleration, clipped to the declared bound, for 2 s and then their speed; followers use the nominal model. Forecast errors flip some correct decisions, which is why it gains little on the validated critic.

4.4 Student

- **Inputs** $\phi(y)$: the critic's traffic features without disturbance parameters, plus the measured accelerations of the current and adjacent leaders; with three lanes also the proposed direction (± 1).
- **Labels, round 0**: run the exact gate on 4096 training episodes; at each proposal store $(\phi(y), g)$.
- **Training**: 2×128 tanh network, Huber loss on g clipped to ± 30, 300 epochs; 20% held out.
- **On-policy relabelling** (package C): run the *student* on 2048 fresh episodes, label the proposals *it* visits with exact g, add them, retrain; twice. Reason: the student's own decisions lead to states the teacher never visits (a wrongly rejected change is proposed again from almost the same state).
- **Threshold**: accept iff $\hat{g} > \tau$, $\tau \in \{-2, -1, 0, 1, 2\}$ selected on 300 validation scenarios.
- **Cost**: one network evaluation, 0.25–0.41 ms including the critic.

4.5 Cascade

At a proposal: accept if $\hat{g} \geq \delta$, reject if $\hat{g} \leq -\delta$, otherwise run the *forecast* gate (deployable). Two choices of δ were tested: the conformal quantile of $|\hat{g} - g|$ on teacher labels (v6; it covers less than claimed, see below), and $\delta \in \{0.5, 1, 2, 4\}$ selected on validation (on-policy version, v7). The on-policy cascade simulates 0.5 (two lanes) to 0.7 (three lanes) decisions per episode.

4.6 Certificate (deterministic case; Corollary 3 of the paper, with the pathwise bound)

For a fixed rule π (the student), the *charge* of one scenario is $C^\pi = \sum |g_k|$ over the proposals at which π decided differently from $g_k \geq 0$, computed offline with exact rollouts at the proposals π visits. Proposition 3 gives, scenario by scenario, $G(\pi) \geq \max\{G(\pi_b), G(\pi_s)\} - C^\pi$. Run π on $n = 500$ fresh calibration scenarios, take q_α = the $\lceil (n+1)(1-\alpha) \rceil$-th smallest charge; then a new scenario satisfies $G(\pi) \geq \max\{G(\pi_b), G(\pi_s)\} - q_\alpha$ with probability at least $1 - \alpha$. Checked on 500 further scenarios per seed.

Why this way

The earlier per-proposal band (conformal on $|\hat{g} - g|$ from teacher labels) assumed the calibration proposals look like deployment proposals. They do not: the student visits other states. Calibrating whole *scenarios* of the deployed rule avoids the problem, because scenarios are independent.

4.7 Baselines (version D)

Stay; MOBIL (tuned); critics with $\bar{c}_0 = c_0$ and validated $\bar{c}_0$; forecast gate; predictive selection without critic (compare "stay to T_f" with "change, then stay to T_f" by forecast rollouts) at every decision, or at most once per 3 s with a 5 s horizon; budget heuristic (on the $\bar{c}_0 = c_0$ critic only).

4.8 Three lanes

Actions stay/left/right; the critic has three outputs; the student also sees the proposed direction; the scenario families include "faster lane two changes away behind a slower middle lane". Everything else is unchanged; Proposition 3 holds because the gate always compares the proposed lane with staying.

5 Version S: stochastic declared model (packages S, T, V, W)

5.1 Traffic

- **Leaders** track a random target speed: $a = \mathrm{clip}(\kappa(v_{\mathrm{tgt}} - v),\ -A|\omega|,\ A|\omega|)$, $\kappa = 1\,\mathrm{s}^{-1}$. The target is piecewise constant on segments of $U(0.5, 3)$ s: the sinusoid's value at the segment end plus noise $0.6A\,U(-1, 1)$, or, with probability 0.2, a hard-braking target $\max(0, \bar{v} - 1.5A - 2)$. The acceleration never leaves the declared bound.
- **Followers**: optimal velocity model with gain $\alpha \sim U(0.5, 1.1)$ per lane and desired speed $U(15, 20)$, both *hidden* from the controller.
- **Safety**: the filter's follower bound is the worst case over the declared ranges ($\alpha \in [0.5, 1.1]$, desired speed ≤ 20 m/s), so Proposition 1 holds for every realization. In tests here: no collision, minimum clearance 2.000 m.
- **Test realizations** are fixed per scenario (seeded by the scenario), so all rules face exactly the same traffic: comparisons stay paired.

5.2 Two simulation models

	Privileged (training only)	**Deployable** (run time)
follower parameters	true values	gain $U(0.5, 1.1)$; desired speed $\max\{U(15, 20),\, v_f + 1\}$ (a heuristic law, which can exceed the declared 20 m/s by up to 1 m/s)
current leader segment	true target and end time kept	target inferred from the measured acceleration (exact unless saturated); end time sampled
later segments	resampled from the true generator	sampled from a prior centred on the current speed
used for	teacher labels, privileged reference gate	queries, Monte-Carlo gates, full rollout

Three follower laws. (1) Version D: scenario-specific follower parameters from the family generators (Table 1; gain 0.8, desired speed $U(15, 20)$). (2) Version S, executed traffic: gain $U(0.5, 1.1)$ per lane and desired speed $U(15, 20)$, hidden; the filter's follower bound is the worst case over these declared ranges, so the safety guarantee concerns this law. (3) Version S, query rollouts: the law q_h of step 5 below, which is history-dependent (desired speed at least $v_f + 1$) and is not the uniform prior; it affects only the simulated outcomes the query averages, which Proposition 5 covers with $\mathrm{TV}(q_h, p(\cdot \mid h))$, never the executed traffic. The sampler tested here is this heuristic q_h, not an exact or validated posterior sampler, which was not tested.

Random streams and the information boundary. Executed traffic: each test or calibration scenario's leader schedule is drawn from a stream seeded by the scenario's parameters (`param_seed`), identical for every rule. Query rollouts: the stream of replicate m is `default_rng([s, m, salt, 29])` with s a hash of the time, the ego's position, speed and lane only (`_state_seed`), so the seed carries no hidden information; the salt separates the streams: 0 deployable queries, 1 privileged training labels and the audit reference (whose first four samples are the labels), 2 the privileged reference gate. Within a query both branches share the replicate's stream (common random numbers). What each query may read is fixed by its kind: the deployable query reads the ego's observations, the leaders' measured accelerations and the declared bounds; the privileged kinds also read the hidden parameters and the current leader segment.

Horizon coverage. A leader schedule has 48 slots; with segments of at least 0.5 s the 47 segments a query draws after the current one cover at least 23.5 s, more than the 15 s horizon.

The deployable sampler exactly (`stoch.py`, `resample`, kind `mc`). For each replicate m of a query at time t, with a random stream seeded by the source state, m and a salt (common random numbers across the two branches), and for each lane k with leader speed v_k, measured acceleration a_k and declared bound $b_k = A_k|\omega_k|$:

1. *Current target.* If the leader is saturated ($|a_k| \geq b_k$): $v_0^{\mathrm{tgt}} = v_k + \mathrm{sign}(a_k)\,(b_k/\kappa + U(0, 3))$; otherwise $v_0^{\mathrm{tgt}} = v_k + a_k/\kappa$ (exact, $\kappa = 1\,\mathrm{s}^{-1}$). Clipped to $[0, v_{\max}]$.
2. *End of the current segment.* $t + U(0, 3)$ s (the elapsed time in the segment is not used).
3. *Generator parameters.* Mean $\hat{\bar{v}} = v_k$; amplitude $\hat{A} = \min\{\mathrm{clip}(b_k/0.45; [0, 5]),\, v_k,\, v_{\max} - v_k\}$ (0 if $v_k = 0$); frequency $\hat{\omega} \sim U(0.2, 0.7)$; phase $\hat{\phi} \sim U(0, 2\pi)$. The true generator's mean, amplitude, frequency and phase come from the scenario family (Table 1; $\omega \in [0.2, 0.8]$).
4. *Later segments* ($K-1$ of them): durations $U(0.5, 3)$ s; target at each segment end $\hat{\bar{v}} + \hat{A}\sin(\hat{\omega} t_{\mathrm{end}} + \hat{\phi}) + 0.6\nu\hat{A}\,U(-1, 1)$, replaced with probability $\min\{0.2\nu, 0.6\}$ by the hard-braking target $\max(0, \hat{\bar{v}} - 1.5\hat{A} - 2)$; clipped to $[0, v_{\max}]$. The leader then tracks the target with acceleration clipped to $\pm b_k$, as in the true model.
5. *Followers.* Gain $U(0.5, 1.1)$; desired speed $\max\{U(15, 20),\, v_f + 1\}$.

The privileged sampler keeps the true current target, segment end and follower parameters and draws later segments from the leader's true generator $(\bar{v}, A, \omega, \phi)$; package Z (Section 12) swaps these parts one at a time.

Neither model knows the realized future: the privileged model knows the hidden *parameters*, not the dice. Both branches of a comparison use the same random numbers (common random numbers), so the noise that affects both branches cancels in g. One sampled advantage still has a standard deviation of about 16 (rare hard-braking events can end in failures), so one sample is a noisy estimate of the sign of g, not an exact sign oracle (the audit of package Y, Section 14, compares one-sample decisions with the student's).

5.3 Teacher labels

Algorithm 2 Labelling one proposal (version S)

1: $\bar{g} \leftarrow$ mean of $K = 4$ *privileged* sampled advantages (both branches to T_f, common random numbers)
2: $a_q \leftarrow$ decision of a deployable *query* with $M_q = 4$ samples: accept iff its mean ≥ 0
3: store $\phi(y)$ (26 lane features + 3 measured accelerations + direction), $\bar{g}$, a_q, and the query's simulated steps

Round 0: the privileged gate acts ($\bar{g} \geq 0$) on 2048 training episodes. Round 1 (on-policy): the student acts on 1024 further episodes and the proposals it visits are labelled the same way.

5.4 Student (mean)

The same network, now trained by *squared error* on the label $\bar{g}$ clipped to ± 60, so its population target is $\mathbb{E}[\mathrm{clip}(\bar{g}; \pm 60) \mid o^+]$, an approximation of $\mathbb{E}[g \mid o^+]$: 14.5% of the labels are clipped with two lanes and 10.9% with three (up to 36% and 22% at a level), with a mean discarded tail $\mathbb{E}[(|\bar{g}| - 60)_+]$ of 5.1 and 2.8. Clipping keeps the sign of each label but pulls large ones toward zero. Rule: accept iff $\hat{g} > \tau$, $\tau \in \{-1, 0, 1\}$ on validation, a further departure from the Bayes sign rule. Lemma 1 shows that the sign of $\mathbb{E}[g \mid o^+]$ is the best decision that uses only o^+; the implemented student approximates it.

5.5 Router

From the same labels, a second network with four outputs predicts

$$\hat{r}_\mathrm{a} \approx \mathbb{E}[[-g]_+ \mid \phi], \quad \hat{r}_\mathrm{r} \approx \mathbb{E}[[g]_+ \mid \phi], \quad \hat{r}_\mathrm{q} \approx \mathbb{E}[d_\mathrm{q} \mid \phi], \quad \hat{\rho} \approx \text{steps of a query},$$

where $[-g]_+$ and $[g]_+$ are the charges of accepting and rejecting, and d_q is the charge of the query's *actual* decision a_q (a query is not a sign oracle). At a proposal:

$$\text{choose } \arg\min\{\hat{r}_\mathrm{a},\ \hat{r}_\mathrm{r},\ \hat{r}_\mathrm{q} + \lambda\hat{\rho}/1000\}, \qquad \lambda \in \{0, 0.5, 2, 8\}.$$

Accept and reject cost nothing extra; "query" runs the deployable Monte-Carlo comparison. $\min\{\hat{r}_\mathrm{a}, \hat{r}_\mathrm{r}\}$ is the predicted charge of the best fixed decision, $\frac{1}{2}(\mathbb{E}|g| - |\mathbb{E}g|)$, which depends on how *spread out* g is given ϕ; a band around the mean cannot see it.

5.6 Band cascade (baseline for the router)

Query iff $|\hat{g}| < \delta$, $\delta \in \{1, 2, 4, 8\}$ on validation; otherwise accept iff $\hat{g} \geq 0$.

5.7 Queries take time (latency)

A query started at time t returns after $\lceil 0.66\,(T_f - t)/T_f/\Delta \rceil$ samples (the measured time of one comparison, shrinking with the remaining horizon; the M samples are assumed to run in parallel). *Waiting protocol* (`stochhl.py`, `GateS`): a *blocking gate* queries every proposal and stays at every step while its query is pending (new proposals are not considered meanwhile); the *cascade* decides by the student and queries, and so waits, only on proposals with $|\hat{g}| < \delta$; the router waits only on the proposals it routes to the query. When an answer arrives, an accepted change is executed only if π_b still proposes the same change and it is admissible; otherwise the ego stays. Per step k and vehicle:

1. if a query is pending: if it is due, execute its accepted change when π_b proposes the same change and it is admissible, else stay; clear it; if it is not due, stay;
2. otherwise, if π_b proposes an admissible change p: the rule either decides now (privileged gate, student, cascade with $|\hat{g}| \geq \delta$) and executes p iff its decision accepts, or dispatches a query with delay K; with $K = 0$ the answer is used at once, with $K > 0$ the ego stays and the query becomes pending, due at $t + K\Delta$;
3. otherwise follow π_b (stay).

This keeps every rule within the structure of Proposition 3 and Theorem 3. There is no failure veto in version S: failures are already charged in the returns. *Which latency model where:* this shrinking model (up to 7 samples at $t = 0$) is used by the deployable gates of package S and by the gate candidate in the certification of Section 8; the uncertainty sweep and packages W, X, Y use a fixed delay of 0 or 1 sample (and the sensing lags of package Y); package T's delay sweep uses fixed delays of $0, 1, 2, 4, 7$ samples.

5.8 Rules compared (version S)

Name	What it does	Simulates at run time
stay, MOBIL	as before; MOBIL uses the nominal follower model	no
critic $\bar{c}_0 = 2$, 10	π_b with the two training costs	no
critic + acc	$\bar{c}_0 = 10$, also sees the measured leader accelerations (same information as the student)	no
student	accept iff $\hat{g} > \tau$ (on-policy labels); student_r0: teacher labels only	no
band	query iff $\lvert\hat{g}\rvert < \delta$	some proposals
router λ	three-way choice above	some proposals
mc1, mc4, mc16	deployable Monte-Carlo gate with $1, 4, 16$ samples at every proposal, with latency	every proposal
mc4 (no latency)	idealized: instantaneous queries	every proposal
priv16	privileged gate, 16 samples, no latency: reference only (knows hidden parameters)	reference
rollall4	rollout algorithm: at every decision, simulate every admissible option followed by π_b (4 samples), execute the best, with latency	every decision

5.9 Certification against an episode-loss requirement

For each frozen candidate (student, band, four routers, mc4) run $n = 500$ fresh calibration scenarios together with π_b and π_s on the same traffic. The realized loss is $L = [\max\{G(\pi_b), G(\pi_s)\} - G(\pi)]_+$. For a tolerance $\varepsilon \in \{1, 2, 5\}$ count $r =$ scenarios with $L > \varepsilon$ and compute the one-sided Clopper–Pearson bound $U = \mathrm{Beta}^{-1}(1 - \beta/K;\, r+1,\, n-r)$, $\beta = 0.1$, $K = 7$. Every candidate with $U \leq \alpha = 0.1$ satisfies

$\Pr[L > \varepsilon] \le \alpha$, all of them simultaneously with probability at least 0.9. The deployed rule is then the cheapest candidate that qualifies.

5.10 Diagnostics

On held-out labels (no simulation): sign agreement and error of the student; average charge of the student, the query, and each routing rule; and, for each routing rule, the predicted stake $\min\{\hat{r}_a, \hat{r}_r\}$ of the proposals it queries versus those it does not. Seed 0 also records wall-clock time per decision (networks; queries with $1, 4, 16$ samples).

5.11 Jobs in package S (per lane count)

Per seed (10 seeds): three critic jobs; `evalS_main` (labels, student, router, tuning, all non-simulating rules, band, routers, certification, timing); `evalS_mc` (mc1, mc4, mc16, mc4 without latency); `evalS_ref` (priv16, rollall4). 60 jobs in total.

6 Is a teacher with privileged information legitimate?

Yes, and it is an established design. The teacher is used only *offline, in simulation*; the deployed student sees only observable inputs (lane geometry, speeds, measured leader accelerations). The same pattern appears as *learning using privileged information* (Vapnik and Vashist, 2009), *learning by cheating* for driving policies (Chen et al., CoRL 2019), teacher–student training of legged locomotion in simulation (Lee et al., Science Robotics 2020), and asymmetric actor–critic training (Pinto et al., RSS 2018).

- **What the student learns.** Trained by squared error on privileged labels, it approximates $\mathbb{E}[g \mid o^+]$ (exactly: $\mathbb{E}[\mathrm{clip}(\bar{g}; \pm 60) \mid o^+]$, Section 5): the privileged advantage averaged over everything the vehicle cannot observe, under the declared distribution of hidden parameters. By the corollary on hidden information, accepting iff $\mathbb{E}[g \mid \phi] \ge 0$ is the *best possible* accept/reject rule that uses only ϕ. Privileged labels are simply a low-noise way to learn that conditional mean.
- **Why not label with deployable simulations?** Their mean is a different quantity (the advantage under the deployable prior), and each label would carry its own model error. The privileged teacher gives unbiased labels for the true advantage; the averaging over hidden quantities is done by the regression.
- **Is the comparison fair?** The critic is trained in the same simulator, so its rewards also reflect the hidden parameters; no learned component sees privileged inputs at run time. The privileged gate (*priv16*) is reported only as a reference, never as a competitor.
- **Caveats.** (i) The student's knowledge is only as good as the declared distribution of hidden parameters (domain randomization); traffic outside it is a mismatch case. (ii) A run-time planner could estimate hidden parameters online (system identification). The plain prior-sampling gate does not; packages W and W2 (Section 12) test an online follower identification and a leader conditioning on the observed history, neither of which improved the query.

7 Exact sampled safety: what is implemented

Kernel filter (Theorem 1). With braking authority $b = 6$ and the leader class "nonreversing, braking no harder than b_L", the sampled emergency envelope has speeds $e_k = (v - kb\Delta)_+$; $D_\Delta(v, v^+) = \max_t E_\Delta(v, t) - L(v^+, t)$ is evaluated exactly at its breakpoints (sample instants, the leader's stop time) and at the zeros of the relative speed on each piece; the state is in $\mathcal{K}_\Delta$ iff $H^+ \ge D_\Delta$. For a first input

u, $\Psi_\Delta(u)$ is the larger of the prefix clearance loss over $[0, \Delta]$ and the terminal requirement after one step; it is convex and nondecreasing in u, so the largest safe input is found by bisection (16 iterations) whose returned value is always the feasible end of the bracket, and the filter clips the command to [lower bounds, $\min\{u_F, \text{upper bounds}\}$]. Feasibility tests use a tolerance of 10^{-9} m. Outside $\mathcal{K}_\Delta$ the held-input certificate is kept as a best effort. Checked against a brute-force maximization on 400 random states (error $< 10^{-10}$) and on the paper's examples (7.5 mm penalty; prefix deficit 0.009 vs. endpoint 0.005).

Terminal-safe commitments (Theorem 2). Within the terminal-safe set the plans are nested, $\mathcal{F}_{\mathrm{c}} \subseteq \mathcal{F}_2 \subseteq \mathcal{F}_N$ (constant, two-input, general). We implement $\mathcal{F}_{\mathrm{c}}$ and $\mathcal{F}_2$ by grid search (41 and 19^2 plans): a plan is feasible if both lanes' clearances hold *in continuous time* over the commitment (front against the leader's minimal displacement, rear against the follower envelope), speeds stay in $[0, v_{\max}]$, and the target-lane state at the end of the commitment lies in $\mathcal{K}_\Delta$. The continuous-time check is exact: on every sample interval, split at the leader's stopping time, each clearance is a quadratic in time, and its minimum is taken over the interval end points and, when the quadratic is convex, its interior vertex (the quantities used in the proof of Theorem 2); speed is affine on each interval, so its bounds are checked at the samples. A grid point that fails proves nothing about the continuous set: an empty grid means only that this finite search found no plan, and the guard then rejects the change. The feasible plan whose first input is nearest the command is executed as stored (no replanning): u_0 for one sample, then the tail input. In package X the rollouts of the gates use the same filter and guard as the executed vehicle, so their advantages refer to the same closed loop. The held-input guard of the sweep ($I_q^{T_m} \cap I_p^{T_m} \neq \emptyset$) is *not* $\mathcal{F}_{\mathrm{c}}$: it has no terminal condition, so the two are incomparable. The general conic reserve $\mathcal{F}_N$ is not run.

The inclusions can be strict (moved here from the paper). With $\Delta = 0.1$, $T_m = 1$, $b = 6$, $v = 1$ and stationary leaders at excess gap 0.3, no constant input is feasible but the two-input plan $(u_0, u_{\mathrm{tail}}) = (-6, -4/9)$ is; at gap 0.2 only a general plan is, e.g. $(-6, -4, 0, \dots)$. The experiments' held-input guard $I_q^{T_m} \cap I_p^{T_m}$ is incomparable with $\mathcal{F}_{\mathrm{c}}$: at $v = 20$, $H^+ = 18$ against a stationary leader it admits $u = -6$ although $D_\Delta(20, 0) = 33.3$.

Earlier offline study of a two-input reserve (deterministic model, without the terminal condition). In the deterministic model, on 2562 real proposal states: same admissions as the constant guard, stored reserve feasible at every sample, clearances kept, same progress, and the executed input deviates from the controller's command by 0.009 instead of 0.175 m/s^2. Over a broad grid of tight-gap states it admits only 0.04% more states (0.15% with a free switching time). It is not used inside the training of package S.

8 The packages and what each answered

Package	What it ran	What the paper takes from it
A, B, C	Deterministic traffic: critics with $\bar{c}_0 \in \{2, 5, 7, 10, 14, 20\}$, exact gate, forecast gate, student, cascade, predictive selector, MOBIL; 15 seeds (two lanes), 10 (three)	Section V-B: exact simulation is enough; student keeps 75% / 29% of the exact gate's gain; pathwise bound checked in 20 000 episodes
S	Stochastic traffic at $\nu = 1$: teacher labels, student, band cascade, routers, Monte-Carlo gates with measured latency, privileged gate, certification; 10 seeds	Certification counts; router results; decision charges
T	At $\nu = 1$: delay sweep $(0, 1, 2, 4, 7$ steps), critic with the student's inputs, label budget $(1/16 \dots 1)$, shifted test traffic	Section V-B (delay) and V-D (information, labels, shift)
V	The uncertainty sweep $\nu \in \{0, 0.5, 1, 2\}$, 10 seeds per level and 20 at $\nu = 1$: critic, student, cascade, prior-sampling gates (1/4/16 rollouts, with and without delay), privileged gate, validated margin, validated $\bar{c}_0$, 4× labels; 140 jobs per lane count	Table I, Fig. 1, Section V-B: the value of simulation and its crossing, the return decomposition, failure causes
benchmark	Numba-compiled rollout timed on one core for $N \in \{3..80\}$, $T_h \in \{5, 15, 30\}$ s	Section V-C: κ_0, κ_1, rollouts per step
control	Critic with its own acceptances delayed by one step (one seed, 700 scenarios)	The delay cost is the cost of postponing a change
W	Prior-sampling gate given the true follower parameters, or the true leader state, or followers identified online; $\nu \in \{0, 1, 2\}$, 10 seeds	Section V-B "What the privileged gate knows"
W2	Prior-sampling gate with the leaders conditioned on their observable history (deployable)	Same paragraph: the negative result
Z	The grouped leader oracle split into current state only and generator only; $\nu \in \{0, 1, 2\}$, 10 seeds, 84 000 runs	Section V-B: the gap is the leaders' generator
X	The frozen $\nu = 1$ rules (critic, student, cascade, 4-rollout prior-sampling gate, privileged gate) under four filters: held-input (the sweep's), kernel filter, kernel filter with $\mathcal{F}_c$ or $\mathcal{F}_2$ guard; 10 seeds, 280 000 runs	Section V-E "Filters of Theorems 1 and 2" and the abstract
Y	The frozen $\nu = 1$ rules: (i) in traffic at $\nu \in \{0, 0.5, 2\}$; (ii) evaluated on traffic 10/30/50 ms old (stale-state sensitivity); (iii) a decision-charge audit with 256 privileged and 16 deployable rollouts per student proposal; 10 seeds, 465 000 runs	Section V-B (fixed-policy sweep, latency, charge audit), V-C, Table I footnote, abstract

9 Package V: the uncertainty sweep

9.1 The factor

The level ν multiplies the two sources of leader randomness of version S: the target noise is $0.6\nu A\,U(-1, 1)$ and the probability of a hard-braking segment is $\min\{0.2\nu, 0.6\}$. The segment lengths $U(0.5, 3)$ s and the hidden follower parameters are present at every level, including $\nu = 0$. So $\nu = 0$ is "random timing of a deterministic target pattern, hidden followers"; $\nu = 1$ is version S; $\nu = 2$ has 40% hard-braking

segments and twice the target noise. The critic is retrained at each level (its environment changes); the PPO controller is shared.

Why this way

One factor, swept from "almost everything about the leaders is predictable from their pattern" to "the leaders' future is mostly dice", is the cleanest way to move the balance between what a learned rule can know in advance and what only sampling can represent.

9.2 Rules evaluated per seed and level

- `critic` ($\bar{c}_0 = 10$), `stay`.
- `student` (teacher labels plus one on-policy round, accept iff $\hat{g} > \tau$ with validated τ), `student_r0` (teacher labels only).
- `margin`: the critic with a validated margin $\theta \in \{0, 0.5, 1, 2, 4, 8\}$ on its own advantage (change only if $Q(\text{change}) - Q(\text{stay}) > \theta$). This control tests whether the gates' gain is merely caution.
- `band_K1`: the cascade, querying the prior-sampling gate (4 rollouts) with a one-step delay when $|\hat{g}| < \delta$ (validated δ).
- `mc1_K0`, `mc4_K0`, `mc16_K0`: the prior-sampling gate with $1, 4, 16$ rollouts and zero delay (the answer is used in the step it was asked for); `mc4_K1`: 4 rollouts, one-step delay.
- `priv16`: the privileged gate, 16 rollouts with the true hidden state, zero delay: the reference.

At $\nu = 1$ additionally: `Vc0` (critics with $\bar{c}_0 \in \{5, 10, 14\}$, the best on validation reported on test) and `Vbig` (the student with $4\times$ the labels).

9.3 Statistics

Every rule runs the same 7 families $\times$ 100 test scenarios with the same traffic realization (common random numbers across rules). For each seed the per-episode returns are averaged over the 700 scenarios; the paper reports the mean over seeds of (rule $-$ critic), a 95% t-interval over seeds ($t_{0.975,n-1}\, s/\sqrt{n}$), and the number of seeds with a positive difference. *This is the only interval type in the paper.* Because the test panel (scenarios and traffic realizations) is the same for every rule and seed, the interval measures variation across trained policies on that panel, not the uncertainty from drawing new traffic. "Value of simulation" is the same statistic for (prior-sampling gate, 16 rollouts, no delay) $-$ student.

9.4 Return decomposition

The return is $\int (w_d v - w_u u^2) dt - c_0 \cdot \#\text{changes} - C_{\text{fail}} \mathbf{1}[\text{failure}]$ with $c_0 = 2$, $C_{\text{fail}} = 200$. For a rule's gain over the critic we compute, per seed, $-2\,\Delta(\#\text{changes})$ (switching part), $-200\,\Delta(\text{failure rate})$ (failure part), and the remainder (progress net of effort). This matters because 0.1% more failures equals 0.2 return units: failure parts are between -0.2 and $+0.3$ for every gate and level; the student's and the gates' gains are progress and fewer lane changes.

10 Package T: latency, information, labels, shift ($\nu = 1$)

- **Delay sweep.** The prior-sampling gate (4 and 16 rollouts) and the cascade with a fixed delay of $k \in \{0, 1, 2, 4, 7\}$ steps. Protocol (Section 5, waiting protocol): while one of its queries is pending the ego stays (for the cascade only after the proposals it queries); when the answer returns, an

acceptance is executed only if the critic still proposes the same change and it is admissible (late drops are rare, 0.01 per episode).

- **Information.** `critic_acc`: a critic that also sees the measured leader accelerations, i.e. the student's inputs; `student_acc`: the student distilled on top of it; `priv16_acc`: the privileged gate on it. Tests whether the student's gain is only one of inputs.
- **Label budget.** Students trained with $1/16, 1/8, 1/4, 1/2, 1$ of the teacher labels (plus, in package V, $4\times$).
- **Shift.** Models trained at $\nu = 1$ tested on traffic with twice the hard-braking probability and $1.5\times$ the target noise.

11 The compute benchmark and the delay control

Benchmark. The full rollout (PPO network, filter interval computation, OVM followers, stochastic leader targets) compiled with Numba, timed on one core of the ARM laptop for $L \in \{2, 3\}$, seven vehicle counts per lane number ($N \in \{3, 5, 7, 10, 20, 40, 80\}$ with two lanes, $\{4, 5, 7, 10, 20, 40, 80\}$ with three) and $T_h \in \{5, 15, 30\}$ s, 42 timings in all. These timings use the held-input filter; the kernel filter and plan search of package X were timed separately in NumPy (Section 13), not compiled, and no end-to-end compiled timing with them is claimed. Timing protocol (`stochastic_V/code/bench_compiled.py`): one machine (Apple-silicon laptop, macOS 26.4.1, arm64; one process, one core); one untimed warm-up call per configuration triggers compilation; then 200 (for $N \leq 20$) or 50 repetitions timed with `time.perf_counter`, each including the construction of the initial state (negligible against the rollout); wall-clock time per rollout. Python and Numba versions were not recorded with the timings. A least-squares fit of $t = (T_h/\Delta)(\kappa_0 + \kappa_1 N) + t_0$ gives $\kappa_0 = 4.67$ and $4.79\,\mu$s, $\kappa_1 = 0.091$ and $0.097\,\mu$s (two, three lanes; `bench/fit.json`), maximum relative error 12.4% and 9.3%. At $N = 7$, $T_h = 15$: 0.86 ms per rollout, 1.73 ms per pair, 58 pairs per 100 ms step. Hence 16 rollouts = 28 ms, 4 rollouts = 7 ms, a two-step lookahead over $B = 3$ options with 16 rollouts each = 10×16 rollouts = 138 ms = 1.4 cores.

Sample requirement. From the privileged labels ($K = 4$ rollouts per proposal), the per-proposal standard deviation has median 3.8 and root mean square 23.7 (two lanes). Used as an illustrative input (an empirical spread, not a worst-case variance guaranteed for every proposal), the compute corollary with $\varepsilon = 1$ asks $M = 0.455 \cdot 23.7^2 = 256$ rollouts. The decision-weighted requirement is far smaller: with $|\bar{g}|$ estimated by the $K = 4$ mean (corrected for its own noise) and a Gaussian sample mean, the expected charge of a privileged M-rollout query is $0.77, 0.44, 0.22, 0.10$ for $M = 1, 2, 4, 8$ against the student's held-out charge 0.45 (two lanes), so about two rollouts match the student. The two charges are estimated differently (the student's against the noisy 4-rollout reference, the query's by a Gaussian plug-in), so this comparison was indicative; the audit of package Y (Section 14) confirms it: one privileged rollout already matches the student.

Delay control. A critic trained here (seed 0, two lanes, $\nu = 1$) with its own acceptances delayed by one step (no simulation at all) loses 0.41 on 700 scenarios; by two steps 1.45. The prior-sampling gate (4 rollouts) in the same run loses 0.44 from a one-step delay. The delay cost is therefore the cost of postponing a proposed lane change, not of simulating; a gate that blocks while waiting pays it whatever the query.

12 Packages W and W2: the information split

The privileged gate knows the hidden state exactly: the follower parameters (gain, desired speed) and, for each leader, the current target speed and the time its segment ends; future segments are sampled from the true generator. The prior-sampling gate draws the follower parameters from the declared

ranges, infers the current target from the measured acceleration ($v + a/\kappa$ when the leader is off its acceleration bound, otherwise a draw beyond the bound), draws the segment end from $U(0, 3)$ s, and samples future segments from a prior centred on the current speed. Package W gives it one privileged item at a time:

- `mcf16`: true follower parameters, leaders as the prior-sampling gate.
- `mcl16`: true leader state and generator (current target, segment end, and future segments from the true generator with the leader's own pattern parameters $\bar{v}, A, \omega, \phi$), followers from the prior. This oracle changes two things at once.
- `mcid16`, `mcid4`: followers drawn from a posterior identified online. The environment keeps, per follower, a grid log-likelihood over gain $\in \{0.5, \ldots, 1.1\}$ (7 values) and desired speed $\in \{15, \ldots, 20\}$ (6 values) of the follower's observed acceleration (exact sensing; tolerance $0.05\,\mathrm{m/s^2}$), updated every step; the query draws from it. The identification is accurate: mean absolute error 0.06 in the gain against 0.15 for the prior.

Package W2 adds `mcidl16`: the leaders conditioned on everything the ego can observe about them. Observable target whenever the leader is off its bound; the onset of saturation as the signal of a target change (saturation can only start at a change); the elapsed time since the last detected change used, with its measured reliability (60%), to draw the remaining segment time; the target under saturation drawn from the declared mixture (pattern plus noise, or hard braking) conditioned on the leader still being at its bound. A sinusoidal fit of the leaders' pattern and the use of the last observed target under saturation were tried and discarded: our sinusoidal fit did not recover the pattern reliably within the 15 s horizon (an unsuccessful estimator, not a proof of non-identifiability), and the last target is stale once the leader saturates.

Package Z separates the two parts of `mcl16`: `mcls16` receives the true current target and segment end but draws future segments from the prior generator (centred on the current speed, random frequency and phase, as the plain query), and `mclg16` infers the current target and segment end as the plain query but draws future segments from the leader's true generator.

Result. The true follower parameters change little (-0.15 to $+0.11$); identification of the followers changes nothing; the observable-history leader conditioning loses 0.3–0.6 to the plain prior query. The grouped leader oracle recovers the privileged gap. Gains over the plain 16-rollout query of the same seed (95% t-interval, seeds positive):

	$\nu = 0$	$\nu = 1$	$\nu = 2$
two lanes			
current state only (`mcls16`)	$+0.11 \pm 0.16$ (8)	$+0.02 \pm 0.21$ (8)	$+0.14 \pm 0.17$ (7)
generator only (`mclg16`)	$+0.95 \pm 0.21$ (10)	$+0.46 \pm 0.28$ (10)	$+0.25 \pm 0.19$ (10)
both (`mcl16`)	$+0.99 \pm 0.21$ (10)	$+0.53 \pm 0.23$ (10)	$+0.30 \pm 0.22$ (10)
three lanes			
current state only	$+0.06 \pm 0.07$ (8)	$+0.01 \pm 0.05$ (6)	$+0.06 \pm 0.04$ (10)
generator only	$+0.84 \pm 0.12$ (10)	$+0.32 \pm 0.06$ (10)	$+0.13 \pm 0.05$ (9)
both	$+0.89 \pm 0.12$ (10)	$+0.34 \pm 0.07$ (10)	$+0.16 \pm 0.12$ (9)

The generator alone carries 81–96% of the grouped oracle's gain, the current state alone almost nothing (resolved only with three lanes at $\nu = 2$, $+0.06$). With the true generator the query beats the student at every level and lane count ($+0.35$ to $+0.70$, 10 of 10 seeds). The information that matters is therefore the leaders' *pattern* (the mean, amplitude, frequency and phase around which their targets are drawn), which the prior sampler redraws at every query and which the estimators we tried did not recover from a short history; it is not the current target or its timing, as an earlier version of this note concluded from the grouped oracle. In the notation of the paper the pattern parameters are part of the full state y and of the hidden quantities θ of Proposition 5.

13 Package X: the filters of Theorems 1 and 2 in the loop

Design. The critics, students and thresholds of the sweep at $\nu = 1$ (seeds 0–9) are frozen and re-run on the same 700 test scenarios under four filters and guards (Section 7). The held-input run reproduces the sweep exactly, run by run, on both lane counts, which checks that nothing else changed. Diagnostics recorded per run: whether the initial and each visited state lay in $\mathcal{K}_\Delta$ (current lane), the number of steps in which the kernel bound was the binding constraint, the number of proposals the terminal-safe guard rejected although the held-input guard admitted them, the failure cause, the minimum clearances and the return.

Result. No initial state, and apart from one step of one cascade run no state at which an input was applied, left $\mathcal{K}_\Delta$; no vehicle collided. The diagnostic counts states at which the filter applied an input; a state in which no input is admissible ends the run and is not counted. The failures labelled front-only under the kernel filter with the held-input guard are of this kind: each ends a held-input commitment, during which the held input is not kernel-filtered, in a state outside $\mathcal{K}_\Delta$ of the target lane (reproduced for scenario *random* 57: a change at $t = 2.0\,\mathrm{s}$ ends at $t = 3.0\,\mathrm{s}$ outside the kernel and fails there). The terminal condition of Theorem 2 excludes exactly these commitments, which is why both terminal-safe guards remove them. In kernel mode a "front" label means that the kernel interval and the remaining lower bounds do not intersect. The kernel filter cut the failures from 0.37–0.53% to 0.10–0.14% (two lanes) and from 0.30% to 0.03% (three), i.e. by 70–90%, and raised every rule's return by +0.53 to +0.86 (10 of 10 seeds) and $+0.62 \pm 0.44$. The whole gain comes from the saved runs: on the runs that failed under neither filter the return changed by −0.0001 (two lanes) and −0.002 (three) on average; the kernel bound was active in 0.03–0.05 steps per run. The two terminal-safe guards gave identical results, rejected an entry the held-input guard admitted only 2–8 times per rule in 7000 runs, and removed the last front-only failures; the remaining 7 and 2 failures in 7000 critic runs are steps where the front kernel and the rear constraint of a closing follower conflict; no front-only filter removes them (Proposition 2 is the rear-only analogue, which needs a sustainable adversarial acceleration that these OVM followers need not have). Cost in NumPy for one vehicle: kernel filter 1.6–1.9 ms per step (held-input 0.3 ms), plan search 4–5 ms per proposal.

Two lanes, $\nu = 1$, 10 seeds × 700 scenarios per rule

Filter, guard	Rule	Return	Δ vs held-input	fail. %	kernel active/run	changes/run
held-input (Prop. 1)	critic	185.80	—	0.50	0.00	0.57
	student	186.01	—	0.50	0.00	0.50
	cascade	186.10	—	0.47	0.00	0.49
	prior gate, 4	185.82	—	0.53	0.00	0.49
	privileged, 16	186.46	—	0.37	0.00	0.49
kernel (Thm. 1)	critic	186.60	$+0.79 \pm 0.37$ (10/10)	0.13	0.04	0.57
	student	186.79	$+0.78 \pm 0.40$ (10/10)	0.13	0.04	0.50
	cascade	186.77	$+0.68 \pm 0.30$ (10/10)	0.14	0.04	0.49
	prior gate, 4	186.68	$+0.86 \pm 0.37$ (10/10)	0.11	0.05	0.49
	privileged, 16	187.00	$+0.53 \pm 0.22$ (10/10)	0.10	0.04	0.49
kernel + $\mathcal{F}_c$	critic	186.67	$+0.87 \pm 0.37$ (10/10)	0.10	0.04	0.57
	student	186.87	$+0.86 \pm 0.37$ (10/10)	0.10	0.04	0.50
	cascade	186.89	$+0.79 \pm 0.31$ (10/10)	0.10	0.04	0.49
	prior gate, 4	186.70	$+0.89 \pm 0.34$ (10/10)	0.10	0.05	0.49
	privileged, 16	187.00	$+0.53 \pm 0.22$ (10/10)	0.10	0.04	0.49
kernel + $\mathcal{F}_2$	critic	186.67	$+0.87 \pm 0.37$ (10/10)	0.10	0.04	0.57
	student	186.87	$+0.86 \pm 0.37$ (10/10)	0.10	0.04	0.50
	cascade	186.89	$+0.79 \pm 0.31$ (10/10)	0.10	0.04	0.49
	prior gate, 4	186.70	$+0.89 \pm 0.34$ (10/10)	0.10	0.05	0.49
	privileged, 16	187.00	$+0.53 \pm 0.22$ (10/10)	0.10	0.04	0.49

Three lanes, $\nu = 1$, 10 seeds $\times$ 700 scenarios per rule

Filter, guard	Rule	Return	Δ vs held-input	fail. %	kernel active/run	changes/run
held-input (Prop. 1)	critic	193.61	—	0.30	0.00	0.94
	student	193.74	—	0.30	0.00	0.80
	cascade	194.03	—	0.30	0.00	0.79
	prior gate, 4	194.08	—	0.30	0.00	0.76
	privileged, 16	194.44	—	0.30	0.00	0.73
kernel (Thm. 1)	critic	194.24	$+0.62 \pm 0.44$ (6/10)	0.03	0.03	0.94
	student	194.37	$+0.62 \pm 0.44$ (6/10)	0.03	0.03	0.80
	cascade	194.66	$+0.62 \pm 0.44$ (6/10)	0.03	0.03	0.79
	prior gate, 4	194.70	$+0.62 \pm 0.43$ (7/10)	0.03	0.03	0.76
	privileged, 16	195.06	$+0.62 \pm 0.44$ (7/10)	0.03	0.03	0.73
kernel + $\mathcal{F}_c$	critic	194.22	$+0.61 \pm 0.45$ (8/10)	0.03	0.03	0.94
	student	194.36	$+0.62 \pm 0.46$ (8/10)	0.03	0.03	0.80
	cascade	194.64	$+0.60 \pm 0.46$ (7/10)	0.03	0.03	0.79
	prior gate, 4	194.68	$+0.60 \pm 0.45$ (7/10)	0.03	0.03	0.76
	privileged, 16	195.04	$+0.60 \pm 0.45$ (8/10)	0.03	0.03	0.73
kernel + $\mathcal{F}_2$	critic	194.22	$+0.61 \pm 0.45$ (8/10)	0.03	0.03	0.94
	student	194.36	$+0.62 \pm 0.46$ (8/10)	0.03	0.03	0.80
	cascade	194.64	$+0.60 \pm 0.46$ (7/10)	0.03	0.03	0.79
	prior gate, 4	194.68	$+0.60 \pm 0.45$ (7/10)	0.03	0.03	0.76
	privileged, 16	195.04	$+0.60 \pm 0.45$ (8/10)	0.03	0.03	0.73

14 Package Y: frozen policies, sensing latency, charge audit

Fixed-policy sweep. The critic, student and thresholds trained at $\nu = 1$ are evaluated, frozen, on the test scenarios of $\nu = 0, 0.5, 2$ (the $\nu = 1$ column is the sweep's own run, seeds 0–9). This separates the effect of the traffic from that of retraining the critic at each level. Gains over the student of the same seed (95% t-interval, seeds positive):

	$\nu = 0$	$\nu = 0.5$	$\nu = 1$	$\nu = 2$
two lanes				
prior gate, 16	-0.42 ± 0.20 (0)	-0.47 ± 0.14 (0)	-0.10 ± 0.26 (3)	$+0.31 \pm 0.17$ (10)
prior gate, 4	-0.38 ± 0.20 (1)	-0.38 ± 0.13 (0)	-0.19 ± 0.17 (1)	$+0.17 \pm 0.23$ (7)
prior gate, 4, one-step delay	-0.88 ± 0.25 (0)	-1.02 ± 0.29 (0)	-0.98 ± 0.39 (0)	-1.11 ± 0.30 (0)
cascade	$+0.03 \pm 0.09$ (6)	-0.01 ± 0.02 (5)	$+0.09 \pm 0.10$ (8)	$+0.05 \pm 0.11$ (7)
privileged gate	$+0.45 \pm 0.18$ (10)	$+0.24 \pm 0.12$ (10)	$+0.45 \pm 0.26$ (10)	$+0.50 \pm 0.27$ (10)
student − critic	$+0.24 \pm 0.15$ (10)	$+0.27 \pm 0.15$ (10)	$+0.21 \pm 0.17$ (9)	$+0.14 \pm 0.18$ (7)
three lanes				
prior gate, 16	-0.01 ± 0.19 (5)	$+0.12 \pm 0.19$ (7)	$+0.32 \pm 0.17$ (9)	$+0.52 \pm 0.22$ (10)
prior gate, 4	$+0.02 \pm 0.18$ (5)	$+0.21 \pm 0.18$ (7)	$+0.34 \pm 0.18$ (9)	$+0.44 \pm 0.22$ (10)
prior gate, 4, one-step delay	-0.99 ± 0.24 (0)	-1.04 ± 0.42 (0)	-0.97 ± 0.25 (0)	-1.07 ± 0.34 (0)
cascade	$+0.21 \pm 0.13$ (10)	$+0.22 \pm 0.12$ (10)	$+0.29 \pm 0.13$ (10)	$+0.20 \pm 0.12$ (9)
privileged gate	$+0.95 \pm 0.26$ (10)	$+0.78 \pm 0.17$ (10)	$+0.69 \pm 0.16$ (10)	$+0.67 \pm 0.23$ (10)
student − critic	$+0.20 \pm 0.18$ (8)	$+0.16 \pm 0.13$ (8)	$+0.13 \pm 0.14$ (7)	$+0.08 \pm 0.14$ (7)

With the policies frozen the value of the prior-sampling query still rises with ν. With two lanes its point estimate changes sign in $(1, 2]$; the interval at $\nu = 1$ contains zero, so the resolved negative and positive values are at $\nu = 0.5$ and $\nu = 2$ ($+0.31 \pm 0.17$, 10 of 10 seeds), and the sign change is located in $(0.5, 2]$; with three lanes it starts at -0.01 ± 0.19, so no negative value is resolved and the retrained sweep's negative value at $\nu = 0$ may partly reflect retraining. The rise is a property of the traffic, not of retraining.

Sensitivity to the age of the information (stale state). What is simulated: the query evaluates the critic's proposal at the decision instant from the traffic state $\tau = 10$, 30 or 50 ms earlier (a snapshot

1, 3 or 5 of the 10 physics substeps before the sample); the vehicle keeps its lane meanwhile; at the sample the change is executed if the query accepted it and it is admissible in the current state. *What this establishes*: how much the gate loses when its information is τ old. *What it does not establish*: a causal real-time pipeline. A query started at $t_k - \tau$ cannot in general know the proposal $\pi_b(o(t_k))$, which is chosen only at t_k; with three lanes it can be either direction. Admissibility at arrival guarantees that an executed change passes the current safety mask, not that its answer could have been computed in advance. With two lanes the only possible change is known in advance, so starting a query at every admissible epoch (also where the critic will not propose) realizes the simulated timeline at the cost of the extra queries; with three lanes, covering both directions with a shared stay branch takes three simulated branches instead of two (42 instead of 28 ms for 16 rollouts at $N = 7$). A causal protocol (proposal frozen at dispatch, answer discarded if the proposal changes, mismatches recorded) was not run. The 28 ms query of Section 11 uses the held-input filter of Proposition 1. Gain over the critic at $\nu = 1$ (zero lag = the sweep):

lag	0	10 ms	30 ms	50 ms
two lanes (student +0.21)				
prior gate, 4	$+0.01 \pm 0.17$	-0.05 ± 0.19	-0.11 ± 0.28	-0.04 ± 0.37
prior gate, 16	$+0.10 \pm 0.22$	-0.15 ± 0.28	-0.02 ± 0.30	$+0.00 \pm 0.20$
cascade (answer in the same step)	—	$+0.30 \pm 0.16$	$+0.25 \pm 0.14$	$+0.31 \pm 0.16$
privileged gate	$+0.66 \pm 0.28$	$+0.57 \pm 0.25$	$+0.59 \pm 0.28$	$+0.65 \pm 0.28$
three lanes (student +0.13)				
prior gate, 4	$+0.47 \pm 0.11$	$+0.33 \pm 0.13$	$+0.35 \pm 0.12$	$+0.28 \pm 0.19$
prior gate, 16	$+0.45 \pm 0.10$	$+0.28 \pm 0.14$	$+0.38 \pm 0.11$	$+0.35 \pm 0.16$
cascade (answer in the same step)	—	$+0.44 \pm 0.09$	$+0.43 \pm 0.09$	$+0.44 \pm 0.10$
privileged gate	$+0.82 \pm 0.09$	$+0.72 \pm 0.12$	$+0.73 \pm 0.10$	$+0.73 \pm 0.10$

Paired against each gate's own zero-lag run the lag costs 0.05–0.26 (two lanes) and 0.07–0.18 (three), against 0.45–1.51 for a one-step delay. With three lanes the 16-rollout gate beats the student at 30 ms by $+0.25 \pm 0.15$ (10 of 10 seeds). The cascade is insensitive to the lag (its same-step answers match its one-step-delayed ones within 0.04).

Decision-charge audit. The student drives 20 test episodes per family (140 per seed); at each of its proposals the job stores 256 privileged rollout pairs at the true state and 16 deployable ones (1707 and 2828 proposals over 10 seeds, two and three lanes). Charges $|\hat{g}|\mathbf{1}\{d \neq \operatorname{sign} \hat{g}\}$ against references from the first K privileged samples (mean over seeds):

	two lanes				three lanes			
reference K	4	16	64	128	4	16	64	128
student	1.62	1.39	1.41	1.39	2.30	2.25	2.21	2.19
prior gate, 1	3.09	2.63	2.63	2.60	2.15	2.09	2.04	2.04
prior gate, 4	1.67	1.34	1.39	1.34	1.53	1.44	1.39	1.38
prior gate, 16	1.30	1.05	1.08	1.03	1.37	1.27	1.22	1.22

The 4-rollout reference overstates the student's and the gates' charges by 5–26% (the critic's by 1–3%), enough to swap close pairs (with two lanes the student and the 4-rollout gate: 1.62 vs 1.67 against four rollouts, 1.39 vs 1.34 against 128). Privileged queries with m rollouts drawn disjointly from the 128-sample reference have charges $0.61, 0.45, 0.23, 0.12, 0.05$ ($m = 1, 2, 4, 8, 16$; two lanes) and $0.35, 0.25, 0.14, 0.08, 0.04$ (three): one privileged rollout already has a lower charge than the student in 18 of 20 seeds. On these test proposals the student's charge is several times its value on the held-out label split, while the deployable gates' is comparable; yet the student's return is higher with two lanes. The charge is a per-decision diagnostic at a common state, not an ordering of rules. The spread of a privileged sample per proposal has root mean square 25.0 (two lanes) and 12.9 (three).

15 Every number in the paper: definition and source

Quantity	Definition and source file
return gain ΔG	mean over seeds of (rule − critic) per-episode return averaged over the 700 test scenarios; 95% t-interval over seeds; seeds positive. `aggV/summary_V.json`, `aggV/value_of_sim.json`
recovered share	ΔG(rule)/ΔG(privileged gate), means over seeds
decision charge	on the student's own held-out split of the labelled proposals: mean of $\lvert\hat{g}_4\rvert\mathbf{1}[d \neq \operatorname{sign}\hat{g}_4]$ with $\hat{g}_4$ the 4-rollout privileged mean at the true state; biased upward by $(\mathbb{E}\lvert\hat{g}_4\rvert - \lvert g\rvert)/2$, the same for every decision independent of the reference noise, so it is a diagnostic, not an ordering of rules. `bench/recharge.json`
rollouts per episode	total rollouts the rule issued divided by test episodes (`stats`)
cores per step	rollouts × 0.86 ms / 100 ms at $N = 7$, $T_h = 15$
failures	episodes ending with no admissible input (the only termination besides T_f); cause front/rear/joint recorded by the environment (`fcause`); completion rate = 1− failure rate
filter comparison	package X: paired by seed against the held-input run of the same rule; `aggV/xfilt.json`, `backX*/`
hard cut-ins	lane entries that force the new follower to brake harder than $3\,\mathrm{m/s^2}$; per episode and per executed change
value of simulation	ΔG(prior gate, 16, no delay) − ΔG(student) paired by seed
decomposition	-2Δ#changes, -200Δfail, remainder. `aggV/decomp.json`
certification	Clopper–Pearson upper bound on $\Pr[L > \varepsilon]$ at level $1 - \gamma/J$, $n = 500$ calibration scenarios, certified iff $\leq \alpha = 0.1$; counts over seeds
compute constants	least squares on `bench/bench_compiled_mac.json`; `bench/fit.json`
information split	paired difference to `mc16_K0` of the same seed; `backW*/`, `backW*b/`

15.1 Certification counts

Corollary 3 at $\nu = 1$, $n = 500$ calibration scenarios per seed, $\alpha = \gamma = 0.1$, $J = 7$ candidates frozen before calibration; a candidate is certified iff its number k_j of scenarios with loss above ε is at most 35, i.e. iff the Clopper–Pearson bound $U_j = \mathrm{Beta}^{-1}(1 - \gamma/J;\, k_j + 1,\, n - k_j)$ is at most α ($U(35) = 0.0992$, $U(36) = 0.1015$). The tolerances $\varepsilon \in \{1, 2, 5\}$ were fixed before calibration; each tolerance is a separate statement at confidence $1 - \gamma$, not a simultaneous one. Entries: mean k_j over the 10 seeds / mean of the per-seed bounds U_j / seeds certified. Every entry is computed from the per-seed counts, which are listed in Table 2. (Version 9 printed the median of the per-seed bounds next to the mean count, which made the two columns inconsistent; the certification decisions are unchanged.)

	two lanes		three lanes	
candidate	$\varepsilon = 1$	$\varepsilon = 5$	$\varepsilon = 1$	$\varepsilon = 5$
student	23.0 / 0.071 / 10	11.3 / 0.042 / 10	52.3 / 0.138 / 0	26.9 / 0.080 / 8
cascade	21.6 / 0.067 / 9	10.6 / 0.040 / 10	47.9 / 0.128 / 0	24.4 / 0.074 / 10
router, $\lambda = 0$	61.8 / 0.159 / 0	25.4 / 0.077 / 10	107.1 / 0.257 / 0	51.0 / 0.135 / 0
router, $\lambda = 0.5$	49.3 / 0.131 / 1	21.2 / 0.067 / 10	95.4 / 0.232 / 0	46.1 / 0.124 / 0
router, $\lambda = 2$	32.1 / 0.092 / 5	15.1 / 0.051 / 10	68.9 / 0.175 / 0	33.7 / 0.096 / 6
router, $\lambda = 8$	23.9 / 0.073 / 9	12.2 / 0.044 / 10	48.5 / 0.130 / 1	25.0 / 0.075 / 8
prior-sampling gate, 4, measured latency	160.0 / 0.368 / 0	59.6 / 0.155 / 0	218.9 / 0.488 / 0	107.2 / 0.257 / 0

16 All results

Gains are over the critic of the same seed; "(a/b)" is the number of seeds with a positive difference.

Table 2: Per-seed counts k_j (certified iff $k_j \leq 35$).

lanes, ε	candidate	seed $0, \ldots, 9$									
2, 1	student	**20**	**33**	**17**	**24**	**24**	**24**	**23**	**28**	**16**	**21**
2, 1	cascade	**17**	37	**14**	**21**	**26**	**18**	**20**	**24**	**19**	**20**
2, 1	router, $\lambda = 0$	44	72	59	71	64	53	58	70	56	71
2, 1	router, $\lambda = 0.5$	**24**	58	44	58	55	40	49	61	45	59
2, 1	router, $\lambda = 2$	**16**	37	**23**	40	39	**28**	**31**	48	**22**	37
2, 1	router, $\lambda = 8$	**17**	**27**	**19**	**23**	**29**	**23**	**25**	36	**17**	**23**
2, 1	prior-sampling gate, 4	160	173	160	174	165	134	140	168	169	157
2, 5	student	**10**	**14**	**8**	**16**	**9**	**8**	**12**	**13**	**11**	**12**
2, 5	cascade	**10**	**15**	**8**	**11**	**13**	**6**	**11**	**12**	**9**	**11**
2, 5	router, $\lambda = 0$	**21**	**26**	**22**	**28**	**22**	**21**	**27**	**26**	**27**	**34**
2, 5	router, $\lambda = 0.5$	**15**	**23**	**17**	**23**	**20**	**15**	**25**	**25**	**21**	**28**
2, 5	router, $\lambda = 2$	**12**	**12**	**11**	**15**	**14**	**11**	**23**	**23**	**11**	**19**
2, 5	router, $\lambda = 8$	**12**	**11**	**11**	**13**	**12**	**8**	**19**	**15**	**9**	**12**
2, 5	prior-sampling gate, 4	69	64	53	57	59	52	55	62	76	49
3, 1	student	66	65	45	62	46	54	43	60	42	40
3, 1	cascade	52	62	42	58	40	49	42	54	41	39
3, 1	router, $\lambda = 0$	116	122	112	104	111	105	98	89	120	94
3, 1	router, $\lambda = 0.5$	103	108	91	101	102	91	90	79	108	81
3, 1	router, $\lambda = 2$	81	89	56	75	67	72	69	53	70	57
3, 1	router, $\lambda = 8$	64	58	**31**	63	48	51	42	50	40	38
3, 1	prior-sampling gate, 4	200	235	220	217	233	218	223	193	239	211
3, 5	student	36	36	**24**	**32**	**23**	**28**	**16**	**34**	**23**	**17**
3, 5	cascade	**30**	**32**	**20**	**33**	**18**	**27**	**17**	**28**	**21**	**18**
3, 5	router, $\lambda = 0$	65	54	55	58	54	50	41	39	49	45
3, 5	router, $\lambda = 0.5$	58	47	44	57	50	44	39	39	47	36
3, 5	router, $\lambda = 2$	43	41	**26**	43	**30**	37	**25**	**29**	**34**	**29**
3, 5	router, $\lambda = 8$	36	**31**	**14**	38	**22**	**26**	**14**	**26**	**21**	**22**
3, 5	prior-sampling gate, 4	110	109	109	118	126	102	103	91	110	94

Bold: certified. The full export (seed, candidate, ε, n, J, γ, k, U, pass) is in `cert_perseed.csv`.

Two lanes

rule	$\nu = 0$	$\nu = 0.5$	$\nu = 1$	$\nu = 2$
critic, mean return	191.7	189.5	185.7	177.5
critic, failures (%)	0.11	0.10	0.49	1.13
critic, lane changes/episode	0.71	0.62	0.57	0.52
privileged gate (16 rollouts)	$+1.10 \pm 0.61$ (10/10)	$+0.58 \pm 0.20$ (10/10)	$+0.63 \pm 0.15$ (20/20)	$+0.63 \pm 0.33$ (10/10)
prior-sampling gate, 16, no delay	$+0.06 \pm 0.44$ (4/10)	-0.22 ± 0.19 (2/10)	$+0.11 \pm 0.14$ (12/20)	$+0.34 \pm 0.19$ (10/10)
prior-sampling gate, 4, no delay	$+0.10 \pm 0.36$ (6/10)	-0.11 ± 0.16 (2/10)	-0.01 ± 0.12 (10/20)	$+0.13 \pm 0.32$ (7/10)
prior-sampling gate, 1, no delay	-0.07 ± 0.48 (2/10)	-0.11 ± 0.18 (4/10)	-0.12 ± 0.17 (6/20)	$+0.08 \pm 0.36$ (6/10)
prior-sampling gate, 4, 1-step delay	-0.35 ± 0.53 (3/10)	-0.83 ± 0.20 (0/10)	-0.73 ± 0.22 (1/20)	-0.96 ± 0.35 (1/10)
student	$+0.67 \pm 0.50$ (10/10)	$+0.28 \pm 0.13$ (10/10)	$+0.24 \pm 0.09$ (18/20)	$+0.13 \pm 0.11$ (9/10)
student, teacher labels only	$+0.62 \pm 0.46$ (10/10)	$+0.27 \pm 0.14$ (10/10)	$+0.26 \pm 0.08$ (20/20)	$+0.25 \pm 0.22$ (10/10)
cascade, 1-step delay	$+0.65 \pm 0.48$ (10/10)	$+0.29 \pm 0.12$ (10/10)	$+0.32 \pm 0.08$ (20/20)	$+0.22 \pm 0.20$ (10/10)
critic + validated margin	-0.28 ± 0.47 (1/10)	-0.04 ± 0.09 (1/10)	-0.06 ± 0.06 (1/20)	-0.07 ± 0.13 (0/10)
always stay	-17.36 ± 0.81 (0/10)	-20.64 ± 0.81 (0/10)	-23.46 ± 0.75 (0/20)	-31.61 ± 3.09 (0/10)
value of simulation (16, no delay − student)	-0.60 ± 0.23 (0/10)	-0.49 ± 0.17 (0/10)	-0.14 ± 0.15 (5/20)	$+0.20 \pm 0.22$ (8/10)
cascade − student	-0.01 ± 0.04 (4/10)	$+0.01 \pm 0.02$ (8/10)	$+0.07 \pm 0.07$ (15/20)	$+0.08 \pm 0.12$ (6/10)
student: progress / switch / fail	+0.27/+0.36/+0.03	+0.07/+0.21/-0.00	+0.08/+0.16/-0.00	+0.04/+0.09/-0.00
prior gate 16: progress / switch / fail	-0.50/+0.47/+0.09	-0.38/+0.28/-0.11	-0.10/+0.19/+0.01	+0.12/+0.11/+0.11
delayed gate 4: progress / switch / fail	-0.92/+0.48/+0.09	-0.93/+0.30/-0.20	-0.73/+0.21/-0.21	-1.10/+0.14/+0.00

Three lanes

rule	$\nu = 0$	$\nu = 0.5$	$\nu = 1$	$\nu = 2$
critic, mean return	199.1	196.6	193.8	188.0
critic, failures (%)	0.04	0.19	0.31	0.57
critic, lane changes/episode	0.93	1.05	0.94	0.96
privileged gate (16 rollouts)	$+1.00 \pm 0.21$ (10/10)	$+1.23 \pm 0.50$ (10/10)	$+0.83 \pm 0.12$ (20/20)	$+1.04 \pm 0.40$ (10/10)
prior-sampling gate, 16, no delay	$+0.04 \pm 0.19$ (8/10)	$+0.44 \pm 0.52$ (7/10)	$+0.44 \pm 0.11$ (19/20)	$+0.80 \pm 0.38$ (10/10)
prior-sampling gate, 4, no delay	$+0.09 \pm 0.18$ (8/10)	$+0.50 \pm 0.39$ (9/10)	$+0.44 \pm 0.12$ (19/20)	$+0.70 \pm 0.29$ (10/10)
prior-sampling gate, 1, no delay	$+0.01 \pm 0.15$ (4/10)	$+0.36 \pm 0.50$ (9/10)	$+0.30 \pm 0.14$ (17/20)	$+0.52 \pm 0.32$ (10/10)
prior-sampling gate, 4, 1-step delay	-1.04 ± 0.32 (0/10)	-0.71 ± 0.65 (1/10)	-0.86 ± 0.18 (0/20)	-0.80 ± 0.44 (2/10)
student	$+0.18 \pm 0.15$ (9/10)	$+0.34 \pm 0.37$ (8/10)	$+0.18 \pm 0.11$ (15/20)	$+0.28 \pm 0.22$ (9/10)
student, teacher labels only	$+0.06 \pm 0.13$ (7/10)	$+0.37 \pm 0.35$ (8/10)	$+0.13 \pm 0.10$ (15/20)	$+0.24 \pm 0.20$ (8/10)
cascade, 1-step delay	$+0.25 \pm 0.17$ (9/10)	$+0.63 \pm 0.46$ (10/10)	$+0.37 \pm 0.11$ (18/20)	$+0.48 \pm 0.29$ (10/10)
critic + validated margin	-0.09 ± 0.19 (0/10)	$+0.02 \pm 0.06$ (1/10)	$+0.00 \pm 0.00$ (0/20)	$+0.00 \pm 0.00$ (0/10)
always stay	-23.83 ± 0.91 (0/10)	-26.77 ± 0.85 (0/10)	-30.43 ± 0.84 (0/20)	-37.86 ± 1.99 (0/10)
value of simulation (16, no delay − student)	-0.14 ± 0.11 (1/10)	$+0.10 \pm 0.23$ (6/10)	$+0.27 \pm 0.10$ (18/20)	$+0.52 \pm 0.25$ (10/10)
cascade − student	$+0.07 \pm 0.05$ (8/10)	$+0.29 \pm 0.12$ (10/10)	$+0.19 \pm 0.08$ (19/20)	$+0.20 \pm 0.14$ (10/10)
student: progress / switch / fail	-0.08/+0.24/+0.03	-0.06/+0.34/+0.06	-0.10/+0.26/+0.01	-0.05/+0.27/+0.06
prior gate 16: progress / switch / fail	-0.44/+0.45/+0.03	-0.15/+0.59/+0.00	+0.06/+0.37/+0.01	+0.31/+0.37/+0.11
delayed gate 4: progress / switch / fail	-1.58/+0.49/+0.06	-1.31/+0.63/-0.03	-1.26/+0.43/-0.03	-1.29/+0.43/+0.06

Information split (paired differences to the prior-sampling gate with 16 rollouts; 10 seeds)

lanes	query	$\nu = 0$	$\nu = 1$	$\nu = 2$
two	true follower parameters	$+0.10 \pm 0.12$ (8/10)	-0.15 ± 0.19 (5/10)	-0.01 ± 0.02 (4/10)
two	true leader state and generator (target, segment end, pattern)	$+0.99 \pm 0.21$ (10/10)	$+0.53 \pm 0.23$ (10/10)	$+0.30 \pm 0.22$ (10/10)
two	identified follower parameters	-0.01 ± 0.05 (5/10)	-0.03 ± 0.14 (5/10)	$+0.02 \pm 0.08$ (6/10)
two	observable leader history (deployable)	-0.35 ± 0.18 (1/10)	-0.61 ± 0.28 (0/10)	-0.41 ± 0.33 (1/10)
two	privileged gate	$+1.04 \pm 0.25$ (10/10)	$+0.56 \pm 0.28$ (10/10)	$+0.29 \pm 0.21$ (10/10)
two	student	$+0.61 \pm 0.23$ (10/10)	$+0.10 \pm 0.26$ (7/10)	-0.20 ± 0.22 (2/10)
three	true follower parameters	$+0.11 \pm 0.03$ (10/10)	$+0.00 \pm 0.04$ (5/10)	$+0.06 \pm 0.05$ (9/10)
three	true leader state and generator (target, segment end, pattern)	$+0.89 \pm 0.12$ (10/10)	$+0.34 \pm 0.07$ (10/10)	$+0.16 \pm 0.12$ (9/10)
three	identified follower parameters	$+0.04 \pm 0.05$ (7/10)	$+0.04 \pm 0.04$ (8/10)	-0.01 ± 0.07 (7/10)
three	observable leader history (deployable)	-0.27 ± 0.12 (0/10)	-0.27 ± 0.17 (0/10)	-0.43 ± 0.18 (0/10)
three	privileged gate	$+0.96 \pm 0.10$ (10/10)	$+0.38 \pm 0.08$ (10/10)	$+0.24 \pm 0.08$ (10/10)
three	student	$+0.14 \pm 0.11$ (9/10)	-0.32 ± 0.17 (1/10)	-0.52 ± 0.25 (0/10)

Package T at $\nu = 1$ (10 seeds): gain over the base critic, mean $\pm$ one standard error, seeds positive

rule	two lanes	three lanes
delay sweep: critic	$+0.00 \pm 0.00$ (0/10)	$+0.00 \pm 0.00$ (0/10)
delay sweep: student	$+0.21 \pm 0.08$ (9/10)	$+0.13 \pm 0.06$ (7/10)
delay sweep: mc4_K0	$+0.01 \pm 0.08$ (5/10)	$+0.47 \pm 0.05$ (10/10)
delay sweep: band_K0	$+0.26 \pm 0.06$ (10/10)	$+0.33 \pm 0.04$ (10/10)
delay sweep: mc4_K1	-0.77 ± 0.16 (1/10)	-0.84 ± 0.11 (0/10)
delay sweep: band_K1	$+0.30 \pm 0.08$ (10/10)	$+0.31 \pm 0.04$ (10/10)
delay sweep: mc4_K2	-1.80 ± 0.15 (0/10)	-2.47 ± 0.13 (0/10)
delay sweep: band_K2	$+0.26 \pm 0.06$ (9/10)	$+0.26 \pm 0.05$ (9/10)
delay sweep: mc4_K4	-3.37 ± 0.18 (0/10)	-5.12 ± 0.16 (0/10)
delay sweep: band_K4	$+0.16 \pm 0.11$ (7/10)	$+0.21 \pm 0.05$ (9/10)
delay sweep: mc4_K7	-6.11 ± 0.19 (0/10)	-9.72 ± 0.28 (0/10)
delay sweep: band_K7	$+0.14 \pm 0.11$ (7/10)	$+0.16 \pm 0.06$ (8/10)
delay sweep: mc16_K0	$+0.10 \pm 0.10$ (5/10)	$+0.45 \pm 0.04$ (10/10)
delay sweep: mc16_K1	-0.57 ± 0.09 (0/10)	-0.80 ± 0.08 (0/10)
delay sweep: mc16_K4	-3.26 ± 0.19 (0/10)	-4.89 ± 0.17 (0/10)
information: critic	$+0.00 \pm 0.00$ (0/10)	$+0.00 \pm 0.00$ (0/10)
information: critic_acc	$+0.21 \pm 0.11$ (6/10)	$+0.49 \pm 0.33$ (5/10)
information: student_acc	$+0.51 \pm 0.15$ (8/10)	$+0.62 \pm 0.40$ (5/10)
information: student_acc_r0	$+0.55 \pm 0.14$ (8/10)	$+0.57 \pm 0.37$ (6/10)
information: priv16_acc	$+0.80 \pm 0.11$ (10/10)	$+1.36 \pm 0.42$ (9/10)
label budget: critic	$+0.00 \pm 0.00$ (0/10)	$+0.00 \pm 0.00$ (0/10)
label budget: student_f0.0625	$+0.10 \pm 0.09$ (6/10)	-0.29 ± 0.10 (2/10)
label budget: student_f0.125	$+0.16 \pm 0.06$ (9/10)	-0.18 ± 0.10 (4/10)
label budget: student_f0.25	$+0.14 \pm 0.08$ (9/10)	-0.07 ± 0.07 (4/10)
label budget: student_f0.5	$+0.18 \pm 0.07$ (9/10)	$+0.08 \pm 0.06$ (8/10)
label budget: student_f1	$+0.25 \pm 0.06$ (10/10)	$+0.12 \pm 0.06$ (7/10)
shifted test traffic: critic	$+0.00 \pm 0.00$ (0/10)	$+0.00 \pm 0.00$ (0/10)
shifted test traffic: critic_acc	$+0.13 \pm 0.21$ (7/10)	$+0.38 \pm 0.33$ (5/10)
shifted test traffic: student	$+0.13 \pm 0.07$ (7/10)	$+0.06 \pm 0.09$ (7/10)
shifted test traffic: priv16	$+0.71 \pm 0.17$ (10/10)	$+0.73 \pm 0.06$ (10/10)
shifted test traffic: never_change	-31.28 ± 1.22 (0/10)	-37.64 ± 0.93 (0/10)
shifted test traffic: loss_eps	–	–

17 Interpretation

1. **Deterministic traffic.** One rollout is exact, the gate improves the critic in every seed, and the student keeps most of it with two lanes. Simulation is enough whenever it fits into the step; learning only buys run-time cost.

2. **Stochastic traffic, the crossing.** The prior-sampling gate loses to the student at low ν (two lanes: -0.61 and -0.49 at $\nu = 0, 0.5$) and wins at high ν (three lanes: $+0.27$ and $+0.52$ at $\nu = 1, 2$). With three lanes the sign change lies in $(0, 1]$; with two lanes, with retrained critics, only the point estimate changes sign in $(1, 2]$. With the $\nu = 1$ policies frozen (package Y) the rise persists in both lane counts and the two-lane values are resolved negative at $\nu = 0.5$ and positive at $\nu = 2$, so the sign change lies in $(0.5, 2]$; with three lanes the frozen value starts at -0.01 ± 0.19, so the negative value at $\nu = 0$ may partly reflect retraining.

3. **Where the sampler's gap lies.** The prior-sampling gate's gap to the privileged reference is the leaders' target generator, their pattern (packages W and Z): given it, the query beats the student everywhere; given only the current target and segment end, it gains almost nothing; given the follower parameters, it gains at most 0.11; conditioned on everything the ego observes about the leaders, it loses. This locates the sampler's gap; it does not by itself explain why the student does better where it does. The student is a function of its observation and can at best approximate $\mathbb{E}[g \mid o^+]$ (Lemma 1): its privileged labels buy a better observation-based rule, not the hidden state of a new episode, and whether they add anything beyond its inputs is left open by the input-matched control. At high ν the future is mostly the declared randomness, which sampling represents. A belief-conditioned sampler was not tested.

4. **Delay.** For a gate that stays while it waits, one step of delay costs the 4-rollout query 0.45–1.51 (it then loses 0.35–1.04 to the critic), all of it progress; a delayed critic pays about the same in a one-seed check. This is the cost of that waiting policy, not of late answers as such: the cascade uses the same late answers and keeps a gain. Evaluating the proposal on traffic 10–50 ms old (package Y, a stale-state sensitivity study, not a demonstrated real-time pipeline) costs only 0.05–0.26.

5. **Compute.** Compiled, the one-step query fits into the step at $N = 7$ with 16 rollouts; the two-step one does not on one core. Compute is not what separated the rules in this study; information was.

6. **Labels and inputs.** Four times as many labels resolve no further gain at these budgets; a critic given the student's inputs gains less than the student (two lanes), but the student's additional gain over it ($+0.31 \pm 0.32$, $+0.13 \pm 0.25$) is unresolved; the student's gain shrinks under shift while the privileged gate's does not, so what it learned is specific to the training traffic.

7. **Certification** (Section 15.1, 10 seeds per lane count, $J = 7$ candidates, tolerances $\varepsilon \in \{1, 2, 5\}$ fixed before calibration, each a separate statement). With two lanes the student, the cascade and the router at $\lambda = 8$ pass at $\varepsilon = 1$ in 10, 9 and 9 of 10 seeds; with three lanes they pass at $\varepsilon = 5$ in 8, 10 and 8 of 10, and at $\varepsilon = 1$ in at most one. The delayed prior-sampling gate passes nowhere. A pass means: with 90% calibration confidence, the candidate's probability of an episode loss above ε is at most 10% under the calibration distribution; it is not a bound on the loss itself.

8. **Safety.** No collision in 3.4 million runs; 98.81–99.99% of the critic's and gates' runs reach the horizon, the rest end with no admissible input (93% through the front constraint under the held-input filter). Run in the loop, the kernel filter of Theorem 1 removes 70–90% of these failures at no measurable cost in progress; what remains are states at which the front kernel and a closing follower's rear constraint admit no common input (a local statement: it does not show that every

earlier control would reach the same conflict). Three guarantees are distinct: the filter's physical clearance, the expected-return bounds of the gates, and the probability of an episode loss above ε (certification).

18 Which result covers which rule

Rule	Guarantee in the paper	Where it is tested
filter, entry guard (Prop. 1)	clearances in both lanes over the certified window (finite window, not recursive)	every run: 3.4 million without collision
kernel filter (Thm. 1)	front viability for all time from $\mathcal{K}_\Delta$, for the declared leader class	package X: in the loop, 70–90% fewer failures
terminal-safe reserve (Thm. 2)	feasibility within a commitment, front (not rear) continuation after it	package X: $\mathcal{F}_{\mathrm{c}}$ and $\mathcal{F}_2$ by grid search; the general conic $\mathcal{F}_N$ not run
privileged gate	Prop. 3, charged at its sampling errors (16 rollouts are not an exact sign oracle)	reference in every stochastic table
prior-sampling gate	Thm. 3 not instantiated (no certified β, D); charges reported as diagnostics	sweep, delay sweep, information split
student, cascade, router	Cor. 3 (episode-loss certificate)	certification counts; shift test